

A Polynomial Parametrization of $xyz + ts + 1 = 0$

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Abstract

We give a four-parameter polynomial family of integral solutions to $xyz + ts + 1 = 0$. The construction is proved by a polynomial identity. Over every field, its restriction to the locus where $xy \neq 0$ is an exact rational parametrization, with an explicit inverse. We also record the precise integral congruence condition governing membership in the family and cover the degenerate locus $xy = 0$.

1 The polynomial identity

Throughout, a *polynomial parametrization* means a tuple of integer-coefficient polynomials whose substitution makes the defining equation vanish identically. Exhaustivity of integral points is a separate question, addressed below.

Theorem 1 (Four-parameter polynomial family). *Fix $\varepsilon \in \{1, -1\}$. For arbitrary $a, b, m, n \in \mathbb{Z}$, define*

$$\begin{aligned} x &= a, & y &= b, \\ z &= \varepsilon(m + n) + abmn, & t &= \varepsilon + abm, & s &= -\varepsilon - abn. \end{aligned} \tag{1}$$

Then

$$xyz + ts + 1 = 0.$$

Equivalently, the five polynomials in (1) define a polynomial map $\Phi_\varepsilon : \mathbb{A}_{\mathbb{Z}}^4 \rightarrow \mathbb{A}_{\mathbb{Z}}^5$ whose image is contained in the hypersurface $xyz + ts + 1 = 0$.

Proof. Put $q = ab$. Since $\varepsilon^2 = 1$, one has the identity

$$\begin{aligned} (\varepsilon + qm)(\varepsilon + qn) &= \varepsilon^2 + \varepsilon q(m + n) + q^2 mn \\ &= 1 + q(\varepsilon(m + n) + qmn). \end{aligned}$$

Under (1), $xy = q$, $z = \varepsilon(m + n) + qmn$, and $ts = -(\varepsilon + qm)(\varepsilon + qn)$. Therefore

$$\begin{aligned} xyz + ts + 1 &= q(\varepsilon(m + n) + qmn) - (\varepsilon + qm)(\varepsilon + qn) + 1 \\ &= q(\varepsilon(m + n) + qmn) - \left[1 + q(\varepsilon(m + n) + qmn)\right] + 1 \\ &= 0. \end{aligned}$$

This is an identity in $\mathbb{Z}[a, b, m, n]$, so every integral specialization gives an integral solution. $\square$

For $\varepsilon = 1$, the family is

$$(x, y, z, t, s) = (a, b, m + n + abmn, 1 + abm, -1 - abn).$$

For example, $(a, b, m, n) = (1, 1, 1, 1)$ gives $(x, y, z, t, s) = (1, 1, 3, 2, -2)$.

2 Exactness on the nondegenerate locus

Theorem 2 (Explicit inverse over a field). *Let K be a field, fix $\varepsilon \in \{1, -1\}$, and set*

$$X(K) = \{(x, y, z, t, s) \in K^5 : xyz + ts + 1 = 0\}.$$

Then Φ_ε restricts to a bijection

$$(K^\times)^2 \times K^2 \longrightarrow \{(x, y, z, t, s) \in X(K) : xy \neq 0\}.$$

Its inverse is

$$a = x, \quad b = y, \quad m = \frac{t - \varepsilon}{xy}, \quad n = -\frac{s + \varepsilon}{xy}. \quad (2)$$

Consequently, the hypersurface is rational, and (1) is an exact rational parametrization on the principal open set $xy \neq 0$.

Proof. If $a, b \neq 0$, then the point supplied by Theorem 1 has $xy = ab \neq 0$. Conversely, take a point of $X(K)$ with $q := xy \neq 0$ and define a, b, m, n by (2). The formulas for x, y, t, s in (1) are then immediate. For z , direct calculation gives

$$\begin{aligned} \varepsilon(m + n) + qmn &= \frac{\varepsilon(t - \varepsilon - s - \varepsilon) - (t - \varepsilon)(s + \varepsilon)}{q} \\ &= \frac{-ts - 1}{q}. \end{aligned}$$

Because the given point satisfies $qz + ts + 1 = 0$, the last expression is exactly z . Thus (2) is a two-sided inverse. Uniqueness also follows at once: a, b are recovered from x, y , and then m, n are recovered from t, s . Thus the displayed maps are mutually inverse regular maps on the indicated open sets.

For completeness, the defining polynomial is irreducible. Indeed, put $R = K[x, y, t, s]$ and view $xyz + ts + 1$ as an element of $R[z]$. The two coefficients xy and $ts + 1$ have no common nonunit divisor. If the linear polynomial factored in $R[z]$, one factor would have z -degree zero and would therefore divide both coefficients; it would have to be a unit. Hence the hypersurface is irreducible. Its nonempty open set $xy \neq 0$ is therefore dense and is isomorphic to $(K^\times)^2 \times K^2$, proving rationality. $\square$

3 The integral image

Proposition 3 (Exact congruence criterion). *Let $(x, y, z, t, s) \in \mathbb{Z}^5$ satisfy $xyz + ts + 1 = 0$ and suppose $q := xy \neq 0$. For fixed $\varepsilon \in \{1, -1\}$, this point is in $\Phi_\varepsilon(\mathbb{Z}^4)$ if and only if*

$$q \mid (t - \varepsilon) \quad \text{and} \quad q \mid (s + \varepsilon).$$

Proof. Every point produced by (1) satisfies $t - \varepsilon = qm$ and $s + \varepsilon = -qn$, proving necessity. Conversely, the two divisibilities make the parameters in (2) integers. The proof of Theorem 2 then shows that these integral parameters reproduce the given point, including its z -coordinate. $\square$

Proposition 4 (Degenerate integral locus). *All integral solutions with $xy = 0$ are covered by the union $\Phi_1(\mathbb{Z}^4) \cup \Phi_{-1}(\mathbb{Z}^4)$.*

Proof. When $xy = 0$, the equation becomes $ts = -1$. Hence either $(t, s) = (1, -1)$ or $(t, s) = (-1, 1)$, while z is arbitrary. Choose $\varepsilon = t$, set $a = x$, $b = y$, $n = 0$, and $m = \varepsilon z$. Since $ab = 0$ and $\varepsilon^2 = 1$, (1) gives

$$z = \varepsilon m = z, \quad t = \varepsilon, \quad s = -\varepsilon,$$

and recovers the prescribed solution. $\square$